

cloudtorch / develop — Step 1 outline

Build a PCA background basis from the last N snapshots of the (t, y, x, λ) cube. Deliverable of `instructions.md` step 1; consumed by step 2. All PyTorch. No fitting here — this step only produces a *fixed* basis.

1. OBJECTIVE

Produce a compact, differentiable spectral dictionary $\{\boldsymbol{\mu}, \mathbf{V}\}$ — a mean spectrum plus $K = 6$ orthonormal eigenprofiles — that spans the observed variability of the *background* (the quiet-Sun profile lying behind the filament). The eigenprofiles are computed *globally*, pooling *all* pixels of the last N frames (see §5a), so a *single shared* basis serves every pixel. In step 2 the background of any pixel becomes $\mathbf{b} = \boldsymbol{\mu} + \mathbf{c} \mathbf{V}$ with only $K = 6$ free coefficients $\mathbf{c}$; those coefficients replace the fixed, per-spectrum `skglm` background (`bdata = rdata - rexttra`) used today.

2. INPUTS & PREPROCESSING

- Cube data of shape (t, y, x, λ) ; take the **last** N time steps $\rightarrow (N, N_y, N_x, L)$.
- Apply the *same* λ -crop (`[100:500]`) and continuum normalisation ($\div \text{Iqs}$) as the fit, so the basis lives in the fitted data's units.
- Flatten space+time into *one* spectra matrix $\mathbf{X} \in \mathbb{R}^{M \times L}$ — **every pixel of all N frames is a row**, $M = N N_y N_x$ ($= 150,800$ for $N=10$). This is a *global, all-pixel* decomposition, not a per-pixel temporal one. Drop NaN rows; optionally random-subsample M to bound memory.

3. PCA

- Mean $\boldsymbol{\mu} = \frac{1}{M} \sum_m \mathbf{X}_m$; centre $\mathbf{X}_c = \mathbf{X} - \boldsymbol{\mu}$.
- Eigendecompose the $L \times L$ covariance $\mathbf{C} = \mathbf{X}_c^T \mathbf{X}_c / (M-1)$ via `torch.linalg.eigh` (cheaper/stabler than full SVD since $L \approx 400 \ll M$; do it in `float64`, then cast). Sort descending; keep the top K eigenvectors as rows of $\mathbf{V} \in \mathbb{R}^{K \times L}$.
- Choose K from the explained-variance curve, not by guess. For MiHI ($N=10$, $M=150,800$) it shows a sharp elbow after 4 modes (90% of the variance), two further genuine modes (5–6, $\rightarrow 94\%$), then a flat photon-noise floor — so we adopt **$K = 6$** . $\mathbf{V}$ is orthonormal $\Rightarrow$ projection is a single mat-mul; K stays a live slider (`set_n_components`), needing no re-solve.

4. OUTPUTS (INTERFACE FOR STEP 2)

A small module, e.g. `pca_background.py`:

- `fit_background_basis(cube, N, K, ...)` $\rightarrow \mu(L), \mathbf{V}(K, L), \text{expvar}(K)$.
- `reconstruct(c)` $= \boldsymbol{\mu} + \mathbf{c} \mathbf{V}$, $\mathbf{c}$ of shape (n_{pix}, K) — linear, autograd-ready.
- `project(spec)` $= (\text{spec} - \boldsymbol{\mu}) \mathbf{V}^T$ — to *initialise* $\mathbf{c}$ from any observed spectrum.
- Persist to `.pt/.npz` so the basis is computed once and reused.

5. DECISIONS (RESOLVED) AND ONE CAVEAT

- Global, all-pixel basis (chosen).** We pool every pixel of all N frames into one matrix and compute *one* shared set of eigenprofiles used by every pixel — **not** a per-pixel *temporal* PCA (a separate basis from each pixel's own N -point time series). The global route shares information across the field and is very well determined ($M \sim 10^5$ rows); the per-pixel route would give only $\leq N-1$ noisy modes per pixel with no spatial sharing.
- What feeds the PCA (chosen: raw).** We use the raw last- N spectra, which Ivan considers filament-free. The alternative — the already-reconstructed backgrounds (`new_fit - new_extrafit`) — stays a one-line switch in the notebook if the raw frames turn out to be contaminated.
- Remaining caveat:** if the filament were *still present* in the last N frames, the basis would absorb the $\text{H}\alpha$ cloud absorption and, in step 3, the background could reproduce the very feature the two clouds must explain — a degeneracy. This is a reason K is kept small ($= 6$, not 10): fewer background modes are less able to mimic clouds. Final check in step 3 — cloud parameters should stay stable as K goes $4 \rightarrow 6 \rightarrow 8$.

Minor: $N = 10$; keep $\boldsymbol{\mu}$ explicitly (don't force a zero-mean model); `float64` eigensolve $\rightarrow$ `float32` basis.

Resolved: $N = 10$ (raw, filament-free) • global all-pixel PCA • $K = 6$ ($\approx 94\%$ of variance; the tail is a photon-noise floor). Next: step 2 — the $K=6$ -parameter background model on this basis.